Average rate of change

1 The temperature, T, in degrees Celsius, of a body at time t minutes is modelled by:

$$T = 37 + 1.4t - 0.02t^2$$

a Find the initial temperature of the body.

b Find the average change in temperature in the first 5 minutes.

c Find an equation for $\frac{dT}{dt}$, the instantaneous rate of change of the temperature with respect to time.

d Find the instantaneous rate of change of the temperature when $t = 5$.

e Find the time, t, when the instantaneous rate of change of temperature equal to zero.

2 The average price in dollars of a particular share, P, on the Stock Exchange is modelled by:

$$P = \frac{1}{3}x^3 - 8x^2 + 63x + 54$$

Where x is the time in years since it was first listed.

a Find the price of the share when it was first listed on the Stock Exchange.

b Find the average rate of change of price for $4 \leq x \leq 6$, correct to two decimal places.

c Find an equation for $\frac{dP}{dx}$, the instantaneous rate of change of price with respect to time.

d Find the instantaneous rate of change of price with respect to time when $x = 4$.

e Find the time(s), x, when the instantaneous rate of change of price is equal to zero.

3 A colony of bacteria is growing in a petri dish and, after t hours, covers an area in cm^2 given by the formula:

$$A(t) = \frac{2}{25}t^2 + \frac{8}{5}t + 92$$

a Find $A'(t)$

b State the unit that can be use to measure $A'(t)$.

c Find the average growth rate of the bacteria between $t = 6$ and 9 hours.

d Find the value of $A'(t)$ when $t = 6$ hours.

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4 The following equation can be used to determine the heart rate of a person whose heart pumps $6000\ \text{mL}$ of blood per minute:

$$R(v) = 6000v^{-1}$$

a Find the rate of change of heart rate with respect to v, the output per beat.

b What units will $R'(v)$ be measured in?

c Find the heart rate at $v = 80$ mL per beat.

d Find the average rate of change between $v = 80$ mL per beat and 100 mL per beat.

e Find the rate of change at $v = 80$ mL per beat.

Rates of change

5 The electrical resistance, R, of a component at temperature, t, is given by:

$$R = 9 + \frac{t}{17} + \frac{t^2}{108}$$

Find $\frac{dR}{dt}$, the instantaneous rate of increase of resistance with respect to temperature.

6 The volume of gas, V, is related to the pressure, P, by the equation $PV = k$, where k is a constant. Find $\frac{dV}{dP}$, the rate of increase of volume with respect to pressure.

7 The asset value of a corporation is expected to change according to the formula:

$$V = -4x^6 - 5x^5 + 250x^4 + 40\,000$$

a Find the asset growth rate, V'.

b Find the value of V' when $x = 7$.

8 A company's market research department finds that the revenue produced by pricing an item at p dollars is given by the equation:

$$R = -3p^2 + 45p$$

a Find $\frac{dR}{dp}$.

b Calculate $\frac{dR}{dp}$ when $p = 4$.

c Calculate $\frac{dR}{dp}$ when $p = 8$.

d Should the market research team recommend a price of $\$4$ or $\$8$. Explain your answer.

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9 The hourly energy consumption in kWh of a city suburb can be modelled using the equation:

$$y = -\frac{1}{8}x^2 + 50x^{\frac{1}{2}} + 40x^{\frac{1}{3}} + 19$$

a Find $\frac{dy}{dx}$.

b Find the value of $\frac{dy}{dx}$ when $x = 11.4$. Round your answer to the nearest whole number.

c State whether the rate of energy consumption is increasing or decreasing at the time when $x = 11.4$.

10 The tide level at a beach is observed to rise and fall according to the following equation:

$$L = \frac{1}{1000}\left(t^4 - 8t^3 + 425t^2 - 1300t\right)$$

a Find L'

b Find the value of L' when $t = 27$.

c State whether the tide is going in or out when $t = 27$.

11 A species of endangered marsupial is reintroduced to an area of bushland. The scientists monitor the population of the marsupial and the area inhabited by them. They find that the relationship between the population, n, and the area, $A\ \text{m}^2$, can be modelled by the following equation:

$$A\left(n\right) = \frac{n^3}{2} - 410n + 38$$

a Find the expression for the rate of change in the area depending on the population.

b Find the rate of increase of the area when the population is 110.

c Find the rate of increase of the area when the population is 370.

d Is the the rate of increase of the area constant? Explain your answer in context.

12 The volume of a reservoir over time changes according to the equation $V = 1000\left(25 - t\right)^3$.

a Find an expression for rate of change of the volume at time t?

b How much water is flowing out at time $t = 20$?

c For what values of t is $\frac{dV}{dt}$ equal to $-300\,000$?

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13 A probe moving through the solar system uses a solar panel to charge its batteries. The amount of millivolts the panel generates depends on the distance in millions of kilometres x from the sun. This relationship can be described by the equation:

$$f(x) = 4000x^{-1} + 8000x^{-2} + 400$$

a Find $f'(x)$.

b Describe what $f'(x)$ represents physically.

c Find $f'(790)$. Round your answer to four decimal places

14 In an experiment involving mice, the daily population of the mice can be approximated by using the following equation:

$$P = 64x^{\frac{1}{2}} + 80x^{\frac{1}{3}} + 8$$

a Find an expression for the daily growth rate of the population, P'.

b Find the daily increase of mice for the following days. Round your answers down to the previous whole number.

i After 10 days **ii** After 40 days **iii** After 80 days

c Hence, describe the population growth.

15 A spherical balloon, initially flat and without any air, is being inflated such that the radius of the balloon, r cm, after t seconds of inflation is given by $r = 4.5t^{\frac{1}{3}}$.

a Find an equation for $\frac{dr}{dt}$, the instantaneous rate of change of the radius with respect to time.

b As t increases, does r increase more slowly, more quickly or at a constant rate?

c Find an equation for the volume, V, of the balloon in terms of t.

d Find an equation for $\frac{dV}{dt}$, the instantaneous rate of change of volume with respect to time.

e Hence, find the instantaneous rate of change of the volume when $t = 8$.

f Describe the rate of change of the volume of the balloon over time.

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16 The effectiveness of a supplement is given by $\frac{dE}{dg}$, where E is the body's reaction to the supplement and g is the number of grams of the supplement administered to the subject. For a particular supplement the body's reaction can be modelled by:

$$E = \frac{g^2\,(400 - g)}{2}$$

a Find the equation for the effectiveness of this particular supplement $\frac{dE}{dg}$.

b Find the effectiveness of the drug when $g = 30$.

c After how many units of the supplement does the effectiveness of the drug start to decrease?

17 As the sand in a hourglass is poured, the radius, r, of the cone formed by the sand expands according to the rule $r = \frac{3t}{5}$, where t is the time in seconds.

a Given that the sand falls such that the height of the cone is the same as the radius at all times, find an equation for the volume, V, of the cone of sand with respect to time, t.

b Find an equation for $\frac{dV}{dt}$, the rate of change of the volume of the cone of sand with respect to time.

c Hence, calculate the instantaneous rate of change of the volume when $t = 4$. Give an exact answer.

18 A company manufactures and sells x items of a given product. The total cost, C, in dollars of producing x items given by:

$$C = 7x + 120\sqrt{x} + 5000$$

a Find the cost of producing 9 items.

b Find the average cost per item when $x = 900$.

c Find an equation for $\frac{dC}{dx}$, the instantaneous rate of change of cost with respect to items produced.

d Find the instantaneous rate of change of cost with respect to items produced when 900 items are produced.

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19 Environmental scientists are monitoring the development of a blue-green algae bloom in a rural dam. They observe that the surface area (in m^2) of the bloom each day is given by the equation:

$$A = \frac{t^2}{2} + \frac{9t}{10}$$

a How much is the surface area increasing each day?

b Find the daily increase at day 20.

c The scientists are allowed to use chemical treatment when the increase in surface area per day exceeds $38.9\ \text{m}^2$/day. When is the earliest they expect to be able to start treatment?

20 A materials laboratory is developing a new type of sponge. They find that the mass of liquid a cube shaped sponge can hold is related to the side-length of the cube by the following equation:

$$M = x^3 + 18x^2 + 81x$$

a Find M'

b State the unit that can be use to measure M'.

c How much is the mass of liquid a sponge can absorb increasing by when the volume of the sponge is $29.791\ \text{cm}^3$?

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